

3-1 (a) Formulate node-voltage equations for the circuit in Figure P3-1. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

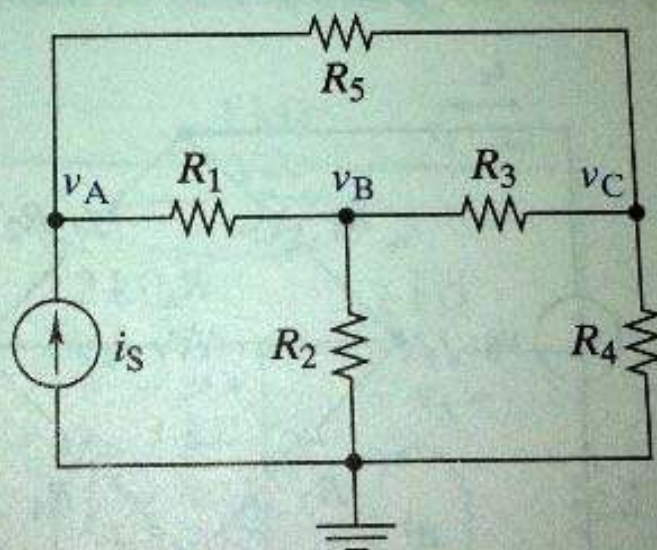


FIGURE P3-1

3-3 (a) Formulate node-voltage equations for the circuit in Figure P3-3. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve these equations for v_A and v_B .
(c) Use these results to find v_x and i_x .

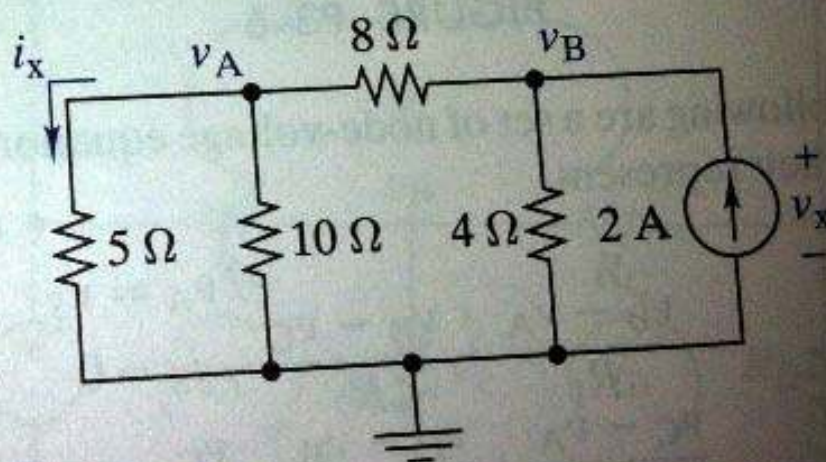


FIGURE P3-3

- 3-4 (a)** Formulate node-voltage equations for the circuit in Figure P3-4.
- (b)** Solve these equations for v_A and v_B .
- (c)** Use these results to find v_x and i_x .

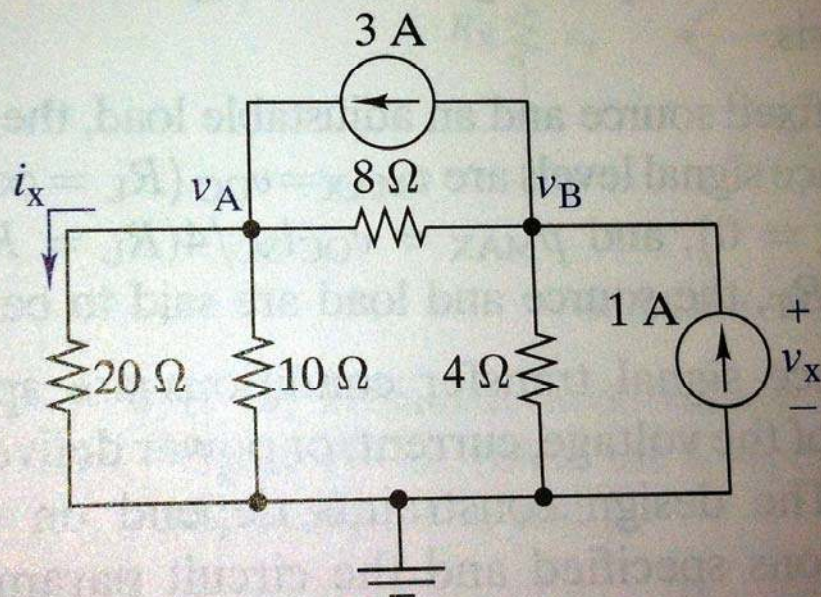


FIGURE P3-4

- 3-5 (a)** Formulate node-voltage equations for the circuit in Figure P3-5. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
- (b)** Solve these equations for v_A and v_C .
- (c)** Use these results to find v_x and i_x .

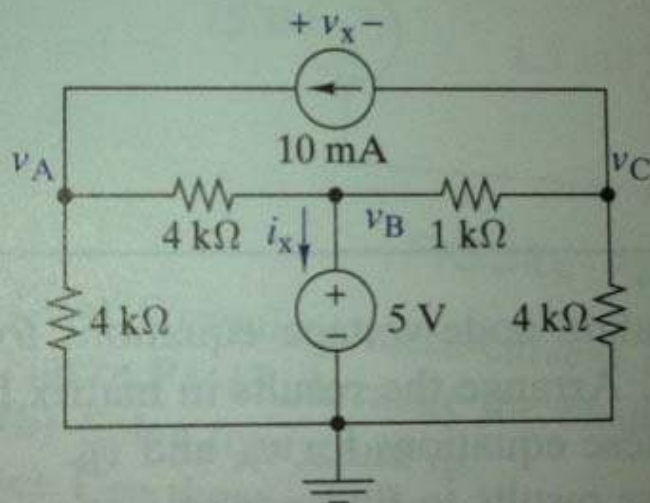


FIGURE P3-5

- 3-13(a)** Formulate mesh-current equations for the circuit in Figure P3-13. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve for i_A and i_B .
(c) Use these results to find v_x and i_x .

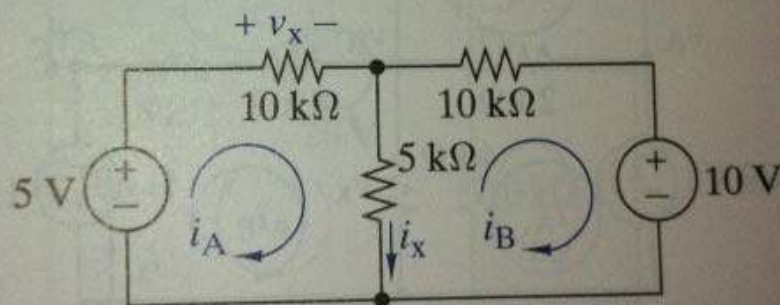


FIGURE P3-13

- 3-17(a)** Formulate mesh-current equations for the circuit in Figure P3-17.
(b) Formulate node-voltage equations for the circuit in Figure P3-17.
(c) Which set of equations would be easier to solve? Why?
(d) Using MATLAB, find v_x and i_x in terms of the mesh-current variables.
(e) Using MATLAB, find v_x and i_x in terms of the node-voltage variables.

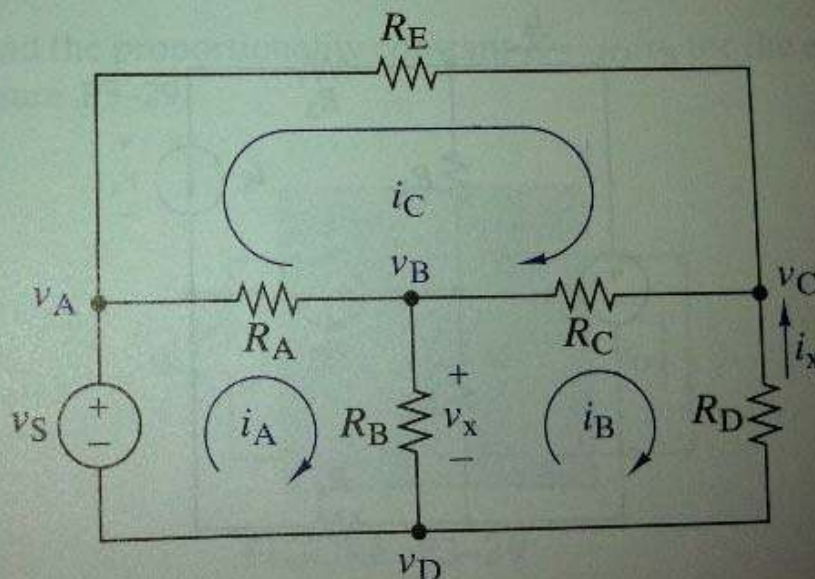


FIGURE P3-17

- 3-20(a)** Formulate mesh-current equations for the circuit in Figure P3-20.
- (b)** Use MATLAB to find symbolic expressions for v_x and i_x in terms of the parameters in the circuit.
- (c)** Find numeric values for v_x and i_x when $R_1 = R_2 = 5.6 \text{ k}\Omega$, $R_3 = 2.2 \text{ k}\Omega$, $R_4 = 1.5 \text{ k}\Omega$, $i_s = 3 \text{ mA}$, $v_{S1} = 15 \text{ V}$, and $v_{S2} = 5 \text{ V}$.
- (d)** Find the power supplied by v_{S1} .
- (e)** Use OrCAD to verify your solutions to parts (c) and (d) are correct.

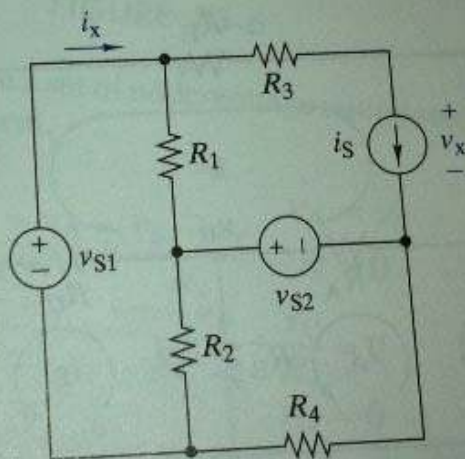


FIGURE P3-20

- 3-22(a)** Formulate mesh-current equations for the circuit in Figure P3-22.
- (b)** Formulate node-voltage equations for the circuit in Figure P3-22.
- (c)** Which set of equations would be easier to solve? Why?
- (d)** Find v_x and i_x using whichever method you prefer.

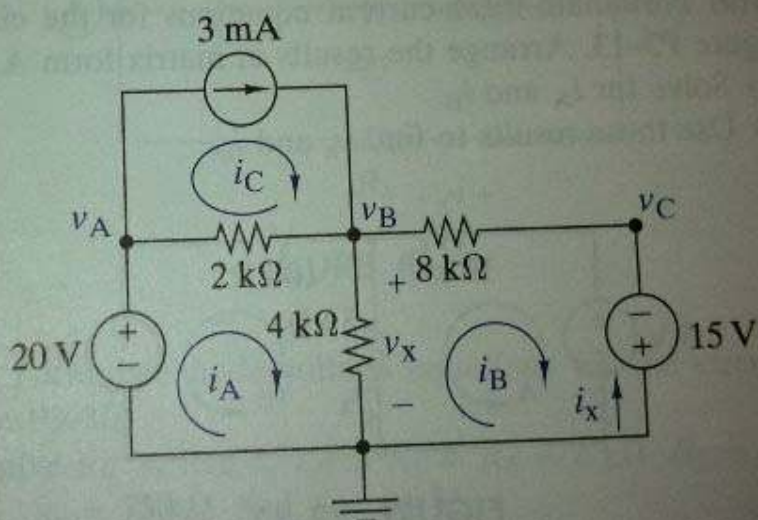


FIGURE P3-22

3-37 Use the superposition principle to find i_O in Figure P3-37. Verify your answer using OrCAD.

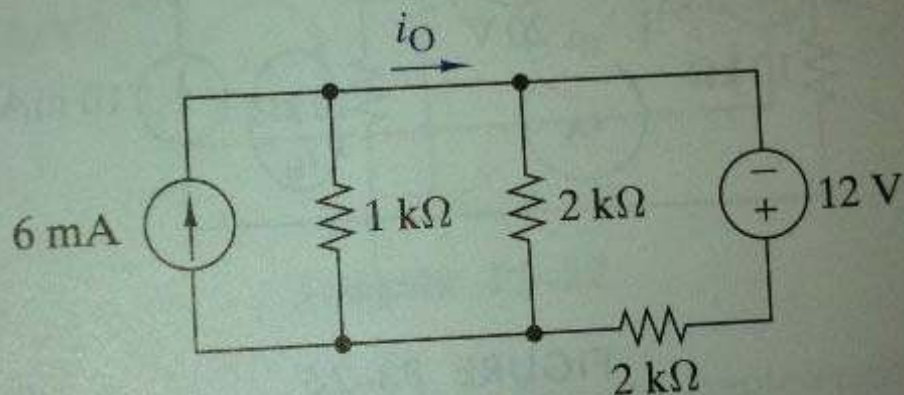


FIGURE P3-37

3-39 Use the superposition principle to find v_O in Figure P3-39.

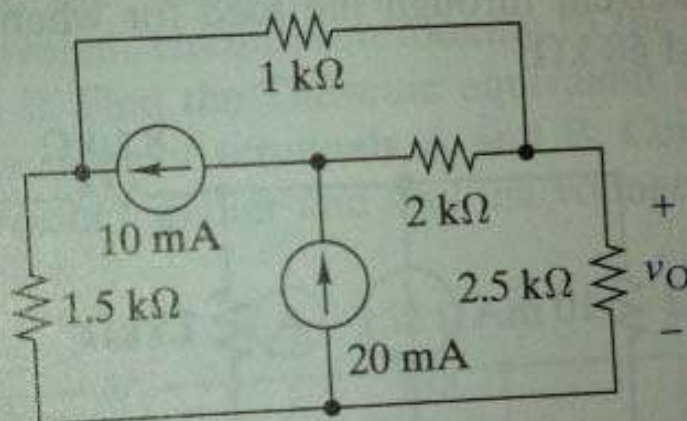


FIGURE P3-39